1 A student is interested in 'bungee jumping', where a person attached to an elastic cord falls from a height and travels downwards through a distance before moving upwards. Different cords are used for different people. A schematic diagram is shown in Fig. 1.1.

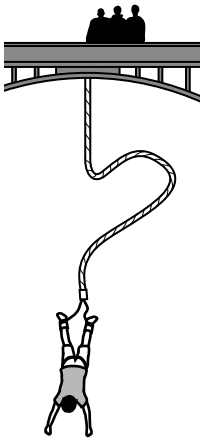

Fig. 1.1

The student models 'bungee jumping' in the laboratory by using elastic cords of unstretched length 50.0 cm with different spring constants. An object is attached to each cord.
The student investigates the relationship between the maximum distance h fallen by the object and the spring constant k of the elastic cord.

It is suggested that the relationship between h and k is

$$\tfrac{1}{2}k(h - L)^2 = mgh$$

where L is the unstretched length of the cord, m is the mass of the object and g is the acceleration of free fall.

Design a laboratory experiment to test the relationship between h and k.
Explain how your results could be used to plot a graph with $\frac{(h - L)^2}{h}$ on the y-axis and to determine the value of g. You should draw a diagram, on page 3, showing the arrangement of your equipment.
In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

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